Algebra II with Trigonometry Chapter 11.3

Olympiad Problems (Gemini)

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Problems

1. Let P be an arbitrary point on the hyperbola $16x^2 - 9y^2 = 144$. Find the product of the perpendicular distances from P to the two asymptotes of the hyperbola.

2. Point P lies on the hyperbola $\frac{x^2}{25} - \frac{y^2}{144} = 1$. If F_1 and F_2 are the foci of the hyperbola and the angle $\angle F_1PF_2$ is exactly 90°, determine the area of $\triangle F_1PF_2$.

3. An ellipse and a hyperbola share the same foci, $F_1(-5, 0)$ and $F_2(5, 0)$. The ellipse has a major axis of length 14, and the hyperbola has a transverse axis of length 8. If the two curves intersect at a point P in the first quadrant, what is the straight-line distance from P to the origin?

4. A circle centered at the origin intersects the hyperbola $x^2 - y^2 = 7$ at four distinct points, which form the vertices of a rectangle. If the area of this rectangle is 48, find the radius of the circle.

5. A hyperbola centered at the origin has its foci on the x-axis. Its asymptotes intersect at the origin to form an angle of 60° that contains the transverse axis. If the distance between the two vertices is 12, find the distance between the two foci.

6. A line tangent to the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ at a point P in the first quadrant intersects the hyperbola's asymptotes at points Q and R. Find the area of the triangle formed by Q, R, and the origin.